

Andrew W. Shenstone

Shenstone, Andrew - Application for Senior Software Engineer, Frontend @ Engine

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SUMMARY

I am applying for Senior Software Engineer, Frontend at Engine because the role matches how I ship: **TypeScript, React**, responsive UI, performance discipline, and a daily **Cursor**/Claude agentic workflow (top 1% Cursor user) while I still own architecture, review, and production risk.

I have 12+ years building customer-facing web and mobile: Bloom admin and commerce integrations, Tilia payments tooling, and founder delivery at Plantfolio (Next.js, accessible components, Stripe). I also have roughly six years on Ruby on Rails through Rails 6 and production React Native (State Fair scale). I collaborate with product and design from scoping through fixes in prod and partner on distributed systems when APIs need it.

Corporate travel and card spend are new domains for me; I am honest about that gap. What transfers is multi-step commerce flows, fintech integrations, finance schooling, and making complex operator UX feel simple.

SKILLS

Frontend (Engine-aligned): TypeScript and React.js in production for 12+ years; Next.js (App Router, RSC) on Plantfolio; long-lived SPAs at Bloom and Tilia, Responsive layouts, cross-browser discipline, and performance-minded delivery (code splitting, loading UX, pragmatic bundle tuning), HTML and CSS depth (Tailwind/Chakra today; adopt house CSS patterns per repo), Accessibility-minded component work; design-system partnership with Figma (Bloom, Tilia Chakra adoption)

Agentic engineering (explicit posting): Cursor (top 1% user): default agentic workflow from scoping through review—not a recent experiment, Claude-class coding agents on

every feature; human ownership of architecture, tests, and production risk, Codebases structured for agent-friendly iteration: typed TypeScript, clear boundaries, Vitest/Cypress

Pluses & full-stack partnership: React Native in production (Ply agency; Washington State Fair scale), Ruby on Rails (~6 years through Rails 6): productive in console and monolith; recent work is React-heavy, REST and GraphQL in production; partner with backend on distributed systems and API contracts, Stripe subscriptions and checkout (Plantfolio); payments-platform tooling (Tilia)

Architecture & async: Monorepos, Async workflows (Inngest, webhooks, background jobs), ETL pipelines, Performance optimization

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

AI-native product & engineering

- Built Plantfolio as an AI-native application: plant ID from photos, contextual assistant (Sage), smart tag extraction, and semantic search over 40,000+ species—LLM features customers use in production, not demos
- AI-native way of building: Cursor (top 1% user) and Claude-class agents are my default workflow from early in the product; I own architecture and review while using AI for speed across the 12+ package monorepo
- Designed the monorepo for agent-friendly iteration—TypeScript, clear boundaries, Vitest/Cypress, and consistent patterns so AI-assisted shipping stays maintainable

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Data modeling & migration: Built Plantfolio on MongoDB and shipped production there for 1+ year, then migrated late to PostgreSQL on Neon (15+ interconnected models with Prisma today)—deep familiarity with document and relational modeling tradeoffs
- Database choices: Evaluated Supabase during exploration but did not ship on it; chose Neon for managed Postgres; auth and subscription tiers via Clerk with Postgres-centric access patterns in production today
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development—with AI-assisted velocity grounded in tests and human review
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)

- Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
 - Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
 - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
 - Role ended in a company-wide layoff affecting approximately one-third of employees
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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
 - Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
 - CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
 - Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus
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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
 - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
 - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
 - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
 - Role ended in a company-wide layoff
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Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects

- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-native plant management platform (vision, chat, RAG search); built with AI-assisted engineering (Cursor-first) and 40,000+ species catalog - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)